

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.3 ROTATION – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

You have been studying the minute hand of the clock on the Administration Block at the University of Nairobi. Answer the driving question: When the minute hand completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up, and how can we use those rules to rotate any shape to any position, on paper and on a grid? Work through the five sections below. Write in full sentences, show every step of your working, and use words and labels from our lessons (centre of rotation, angle of rotation, clockwise, anticlockwise, order of rotational symmetry). Draw neat sketches where a prompt asks for them. Use a ruler, compass and protractor where needed.

### Part 1: What changes and what stays the same?

**The minute hand of the University of Nairobi clock was observed at 12:00, 12:15, 12:30 and 12:45. The hand pointed in a different direction each time, yet it looked the same. (a) Name the type of motion the hand is performing and say how it differs from a translation and from a reflection. (b) Say which point is the centre of rotation, and explain why it is the only point that does not move. (c) List what changes and what stays the same about the hand during the motion, and give a reason for each. (d) Explain how a compass and a protractor in the four-step construction method represent features of the clock.**

(a) The hand is performing a rotation: every point turns about a fixed point. A translation slides the object without turning it, and a reflection flips it over a mirror line. The hand does neither of those. (b) The centre of rotation is the pivot (axle) at the centre of the clock face. It is the only invariant point because it is the point the whole hand turns about, so its distance from itself is zero and it stays where it is. (c) What changes: the direction the hand points and the position of every point except the centre. What stays the same: the length and shape of the hand, the distance of every point from the centre, and the sizes of the angles. The object and image are congruent, because rotation does not stretch, shrink or flip the hand. (d) The compass pin is the fixed centre of the clock. The compass arc is the circular path traced by any point on the hand, so the distance from the centre is kept. The protractor angle is the angle through which the hand turns. Between them, the same four steps (mark the centre, draw rays, compass for distance, protractor for angle) predict where each point lands.

## Part 2: Predicting positions with coordinate rules

**Model the minute hand as a line segment on a coordinate grid with the clock centre at the origin  $O(0, 0)$ . The points  $A(0, 5)$ ,  $B(0, 3)$  and  $C(0, 1)$  lie on the hand at 12:00. The hand turns  $90^\circ$  clockwise every 15 minutes.**

| Time  | Turn from 12:00       | A      | B      | C      |
|-------|-----------------------|--------|--------|--------|
| 12:00 | $0^\circ$             | (0, 5) | (0, 3) | (0, 1) |
| 12:15 | $90^\circ$ clockwise  | ?      | ?      | ?      |
| 12:30 | $180^\circ$           | ?      | ?      | ?      |
| 12:45 | $270^\circ$ clockwise | ?      | ?      | ?      |

**(a) Write down the coordinate rule for each turn:  $90^\circ$  clockwise,  $180^\circ$  and  $270^\circ$  clockwise. (b) Use the rules to complete the table. (c) Using point A at 12:15, show by calculation that its distance from the centre has not changed. (d) State the position**

(a)  $90^\circ$  clockwise:  $(x, y) \rightarrow (y, -x)$ .  $180^\circ$ :  $(x, y) \rightarrow (-x, -y)$ .  $270^\circ$  clockwise (the same as  $90^\circ$  anticlockwise):  $(x, y) \rightarrow (-y, x)$ . (b) 12:15:  $A(5, 0)$ ,  $B(3, 0)$ ,  $C(1, 0)$ . 12:30:  $A(0, -5)$ ,  $B(0, -3)$ ,  $C(0, -1)$ . 12:45:  $A(-5, 0)$ ,  $B(-3, 0)$ ,  $C(-1, 0)$ . Check for A at 12:15 using  $(0, 5) \rightarrow (5, 0)$ . (c) At 12:00, the distance  $OA = \sqrt{0^2 + 5^2} = 5$ . At 12:15,  $OA' = \sqrt{5^2 + 0^2} = 5$ . The distances are equal, so the distance from the centre is preserved. (d) At 1:00 the hand has made a full  $360^\circ$  turn, and the rule for  $360^\circ$  is  $(x, y) \rightarrow (x, y)$ . So A is back at (0, 5), the same position as at 12:00.

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| of A at 1:00 and explain why. |  |
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### Part 3: Finding the centre and angle of rotation

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| <p><b>Suppose you did not know where the clock's pivot was. You are only told that the tip of the hand moved from A(0, 5) at 12:00 to A'(5, 0) at 12:15, and then to A''(0, -5) at 12:30. (a) Describe, step by step, how you would use a ruler and compass to find the centre of rotation from these positions. (b) Plot A, A' and A'' on a coordinate grid and carry out your construction on the grid. Use it to state the coordinates of the centre. (c) Explain why the centre must lie on the perpendicular bisector of each object-image segment. (d) How would you measure the angle of rotation and check your answer?</b></p> | <p>(a) Join A to A' and A' to A''. Construct the perpendicular bisector of each segment with a compass and ruler. The point where the two bisectors intersect is the centre of rotation. (b) On the grid, the perpendicular bisector of AA' is a straight line through the middle of AA' that meets it at a right angle, and the perpendicular bisector of A'A'' is constructed in the same way. The two bisectors cross at the origin, so the centre is O(0, 0), the clock's pivot. (c) Rotation preserves distances, so the centre is the same distance from a point and its image (<math>OA = OA'</math>). Every point equidistant from A and A' lies on the perpendicular bisector of AA'. This is true for every object-image pair, so two bisectors are enough to meet at exactly one point. (d) With the protractor at O, measure the angle between ray OA and ray OA'. It is <math>90^\circ</math>, and it is clockwise because the hand moved from 12 towards 3. To check, rotate A by <math>90^\circ</math> clockwise about O using the compass and protractor (or the rule <math>(x, y) \rightarrow (y, -x)</math>) and confirm that it lands on A'. Do the same with another point to show the whole object maps onto its image.</p> |
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### Part 4: Rotational symmetry

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| <p><b>(a) What does it mean for a plane figure to have rotational symmetry of order n, and how do you work out the smallest angle of rotation that maps it onto itself? (b) State the order of rotational symmetry of a regular hexagon (like the weave pattern of a Maasai kiondo basket), a rectangle, and the letter S. (c) The clock face at the University of Nairobi has 12 equally spaced marks. State its order of rotational symmetry and the angle between each coincident position. (d) A</b></p> | <p>(a) A plane figure has rotational symmetry of order n if it maps onto its own outline exactly n times during one full <math>360^\circ</math> turn about its centre. The smallest angle of rotation is <math>360^\circ \div n</math>. (b) Regular hexagon: order 6 (<math>60^\circ</math> each time). Rectangle: order 2 (<math>180^\circ</math>). Letter S: order 2 (<math>180^\circ</math>). (c) The clock face has order 12, and the angle between positions is <math>360^\circ \div 12 = 30^\circ</math>. (d) A triangular prism has order 3 about its longitudinal axis, because its end faces are equilateral triangles, which map onto themselves 3 times in a full turn (every <math>120^\circ</math>).</p> |
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| triangular prism is rotated about its longitudinal axis. State the order of rotational symmetry it has about that axis and explain why. |  |
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## Part 5: Answering the driving question

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| <p><b>Write a complete, evidence-based answer to the driving question: When the minute hand of the University of Nairobi clock completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up, and how can we use those rules to rotate any shape to any position, on paper and on a grid? Your answer should (i) state the properties of rotation, (ii) explain how to find the image on plain paper and on the Cartesian plane, (iii) explain how to find the centre and angle when they are unknown, and (iv) give one example of rotation from your school, home or a Kenyan place, stating its centre and an angle or order of symmetry. Finally, say how your thinking has changed since your first sketch in Lesson 1.</b></p> | <p>Rotation turns every point of an object through the same angle about one fixed point, the centre of rotation. The centre is the only point that does not move. Every other point stays the same distance from the centre, so the object and its image are congruent: the sides and angles are equal. The direction (clockwise or anticlockwise) must always be stated because it decides where the image lands. On plain paper, I mark the centre, draw a ray through each vertex, use a compass to keep the distance from the centre, and use a protractor to measure the given angle and mark the image point. This works whether the centre is inside, outside or on the boundary of the shape. On the Cartesian plane, rotations about the origin follow rules: <math>90^\circ</math> clockwise <math>(x, y) \rightarrow (y, -x)</math>; <math>180^\circ</math> <math>(x, y) \rightarrow (-x, -y)</math>; <math>270^\circ</math> clockwise <math>(x, y) \rightarrow (-y, x)</math>; <math>360^\circ</math> <math>(x, y) \rightarrow (x, y)</math>. For the clock, the tip A(0, 5) goes to (5, 0) at 12:15, (0, -5) at 12:30, (-5, 0) at 12:45 and back to (0, 5) at 1:00, and its distance from the centre stays 5 every time. If the centre and angle are unknown, I join object points to their images, construct two perpendicular bisectors, and use their intersection as the centre. I then measure the angle at the centre between an object point and its image, and I check by rotating the object. Rotational symmetry shows how often a shape maps onto itself in <math>360^\circ</math>: the clock face has order 12 (<math>30^\circ</math>) and a regular hexagon has order 6 (<math>60^\circ</math>). A real-life example is a ceiling fan with three blades. Its centre of rotation is the motor axle and it has order 3, so it maps onto itself every <math>120^\circ</math>. At the start, I only knew the hand had moved. Now I can describe the motion with exact rules and calculate where every point ends up.</p> |
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| RUBRIC                                            |                                                                                                                                                            |                                                                                             |                                                                                                     |
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| Criterion                                         | Excellent (4)                                                                                                                                              | Proficient (3)                                                                              | Developing (1–2)                                                                                    |
| Properties of rotation and link to the phenomenon | Correctly names the centre, angle, direction and preserved distances and shape, and clearly links each to the Nairobi clock and to the construction steps. | States most properties correctly with some link to the clock; minor gaps in explaining why. | Names rotation but confuses it with translation or reflection, or lists properties without reasons. |

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| Applying coordinate rules            | Uses all rules correctly to complete the table, verifies distance preservation with calculation, and explains the $360^\circ$ return.             | Completes most of the table correctly; minor sign errors or incomplete distance check.    | Uses wrong rules or direction, or cannot calculate image coordinates. |
| Finding centre and angle of rotation | Gives an accurate perpendicular bisector construction, finds the correct centre, justifies why the centre is equidistant, and verifies the angle. | Describes the method and finds the centre with minor errors in reasoning or verification. | Cannot locate the centre or does not use perpendicular bisectors.     |
| Rotational symmetry                  | Gives correct orders and angles for all figures and the solid, with the $360^\circ \div n$ reasoning.                                             | Most orders correct; reasoning is partial.                                                | Orders incorrect or no use of $360^\circ \div n$ .                    |
| Synthesis and real-life application  | Writes a coherent, evidence-based answer covering all four points, gives a valid Kenyan example, and reflects on how their thinking changed.      | Answer covers most points with a reasonable example; reflection is brief.                 | Answer is incomplete or lacks evidence, example or reflection.        |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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